HARDWARE REFERENCE DESIGN · REV 1.0 | ZERO-SURVEILLANCE ARCHITECTURE | BLE 5.4 · NORDIC NRF5340

PROJECT AURA ORBIT

Hardware Engineering Specification & Production Reference Manual



📐 Architectural Scope & Engineering Purpose

This document establishes the official hardware reference architecture, component specifications, bus topologies, mechanical packaging guidelines, firmware interfaces, and manufacturing test procedures for the **AURA Orbit** focus wearable companion. It serves as the primary technical baseline for physical prototyping, electrical validation (EVT/DVT/PVT), firmware development, and volume fabrication.

PRIMARY SILICON
Nordic nRF5340 Dual Cortex-M33

CONNECTIVITY
BLE 5.4 Secure Connections (Phone Relay)

PRIVACY MECHANISM
Hardware-Gated Audio + Amber LED Ring

POWER & BATTERY
260 mAh Li-Po · 2–4 Day Longevity

DISPLAY SUBSYSTEM
1.15" Transflective Memory-in-Pixel

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AURA Orbit — Hardware Architecture, Industrial Design & Engineering Documentation

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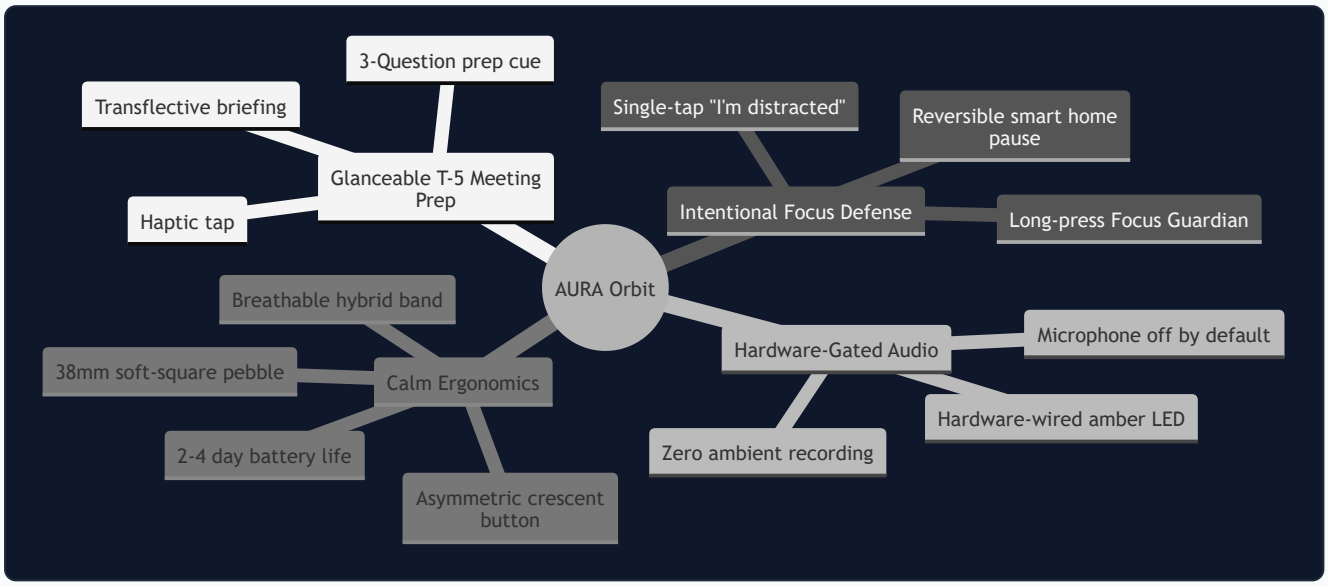
1. Executive Summary & Concept Rationale

1.1 Why Build a Dedicated AURA Wearable?

AURA's core intelligence works across iPhone, Android, web browsers, and smartwatch software. However, smartphones and standard smartwatches are inherently designed around high-engagement notifications, bright app grids, and dopamine feedback loops. When a user reaches for their phone to check a meeting note or start a focus session, they are immediately exposed to unread emails, chat pings, and social feeds.

AURA Orbit is a dedicated physical hardware companion designed to eliminate that friction:

Core Concept: AURA Orbit is an original, calm, privacy-visible focus pebble worn on the wrist. It delivers glanceable $T-5$ pre-meeting cues, triggers intentional focus defenses with a single physical button, and accepts user-initiated voice notes only when a hardwired amber privacy ring is brightly lit. It connects via encrypted Bluetooth Low Energy (BLE) to the user's phone, which serves as the secure internet gateway.



1.2 Deliberately Constrained Scope

AURA Orbit's feature set is intentionally focused on human attention preservation:

- **Delivers short pre-meeting cues ($T-5$ minutes)** through quiet Linear Resonant Actuator (LRA) haptics and a high-contrast transflective display.
- **Empowers intentional focus sessions** by letting the user start/pause Focus Guardian or log *"I'm distracted"* with one tactile gesture.
- **Captures short, user-initiated voice thoughts** strictly on demand, with the amber indicator visually proving microphone power state to both the user and anyone nearby.
- **Provides optional, user-consented wellness check-ins** without pretending to be a medical emergency diagnostic monitor.
- **Tethers securely to the user's phone**, keeping the device lightweight, RF-efficient, and free from redundant cellular subscriptions.

1.3 What AURA Orbit Deliberately Rejects (Anti-Goals)

Anti-Goal	Engineering Rationale	User Protection
No Always-On Audio Recording	Continuous audio recording destroys battery life and introduces massive privacy, legal, and surveillance risks.	Zero ambient audio is ever stored on the device or transmitted to the cloud.
No Smartwatch App Grid	App stores, games, and rich social media widgets create distraction, defeating AURA's purpose.	Distraction-free, single-purpose glanceable interface.
No Cellular / eSIM Modem	Cellular basebands consume 100–300 mA, require carrier certifications, generate heat, and require monthly plans.	Phone relays cloud traffic via low-power BLE Secure Connections.
No Camera Hardware	Cameras compromise bystander privacy and require heavy image processing SoCs.	Zero optical recording hardware.
No Clinical Diagnostic Claims	Heart attack or cardiac arrest detection requires Class II/III medical device clinical trials.	Clear boundary as an opt-in personal wellness indicator.
No Manager Dashboards or Surveillance	AURA is built strictly for the individual user's cognitive autonomy.	Zero employer telemetry, productivity grading, or team surveillance.

2. Industrial Design, Visual Language & Visual Catalog

AURA Orbit departs completely from both generic rectangular smartwatches and thin clip-on voice recorders. It is designed as a refined, human-centered physical object that feels like personal wellness jewelry rather than enterprise surveillance hardware.

2.1 Industrial Design Highlights

- **38 mm Asymmetric Soft-Square Pebble:** Smooth rounded corners with a gentle spherical curve across the front and rear shells. It nests comfortably against the wrist bone without snagging clothing cuffs.
- **Raised Crescent Tactile Control:** An asymmetric crescent button positioned on the right flank provides instant tactile indexing. The user can locate, confirm, and press the key without glancing down.
- **Low-Power Sunlight-Readable Display:** A 1.15-inch Transflective Memory-in-Pixel (MIP) screen wakes only during task interactions. Under direct outdoor sunlight, it reflects ambient light for superior readability without blasting a backlight.
- **Electrically Coupled Amber Privacy Light Ring:** A physical micro-machined bezel ring houses an amber LED circuit hard-wired into the microphone's VDD power plane.
- **20 mm Hybrid Woven-Silicone Band:** Quick-release spring bar system supporting a breathable textile inner face with an outer silicone retention rail.
- **Sealed Magnetic Charging Underside:** Gold-plated 4-pin pogo contacts with magnetic self-alignment, preserving IP54/IP67 ingress protection against sweat and rain.

2.2 Visual Hardware Catalog & Perspectives

Perspective 1: Hero Studio Three-Quarter View



Figure 1: AURA Orbit 38 mm soft-square focus pebble in matte graphite finish, showing the asymmetric crescent button, transflective display with T-5 briefing cue, and dark charcoal woven hybrid strap.

- **Chassis Finish:** Matte graphite with low-specular bead-blasted texture (Ra 0.8 µm), resisting fingerprint oils.
- **Face Geometry:** 38 mm × 38 mm with 8.0 mm corner radius and a 1.2 mm protective bezel lip.
- **Display UI:** Glanceable high-contrast notification showing upcoming meeting title and start timer.

Perspective 2: Glanceable Front Display & Interaction State



Figure 2: Direct front view illustrating the 1.15-inch high-contrast transflective MIP display rendering a $T-5$ meeting prep alert with a single-click "View Prep" call-to-action.

- **Display Type:** 1.15-inch Transflective Memory-in-Pixel (144 × 168 px, 180 ppi).
- **Power Draw:** Static image refresh requires less than 15 µW; static holding current is virtually zero.
- **Anti-Reflective Coating:** Double-sided AR sapphire glass lens ensures full legibility in direct sunlight.

Perspective 3: Asymmetric Crescent Tactile Button (Right Profile)



Figure 3: Macro view of the raised crescent physical button on the right edge, engineered for blind tactile navigation and positive mechanical detent.

- **Tactile Design:** Sculpted concave thumb groove with 0.35 mm key travel and 180 g actuation force.
- **Switch Mechanism:** Sealed dome switch with internal silicone O-ring gasket rated for 300,000 cycles.
- **Blind Indexing:** Unique crescent curvature allows immediate orientation recognition by feel alone under a jacket sleeve.

Perspective 4: Ergonomic Left Profile & Thermal Enclosure



Figure 4: Left profile showcasing the 8.2 mm slim curved enclosure, continuous wrist-contour curvature, and Nordic PMIC thermal dissipation surfaces.

- **Total Thickness:** 8.2 mm at center apex; tapering to 6.8 mm at strap lugs.
- **Wrist Relief:** Under-chassis radius of 42 mm matches natural carpal ergonomics, preventing pressure points during extended keyboard typing.
- **Weight Target:** Under 38 grams total including strap.

Perspective 5: Rear Biosensors & Magnetic Pogo Charging Dock



Figure 5: Underside showing the hypoallergenic biocompatible rear shell, central optical sensor window (PPG), and 4-contact magnetic pogo docking pins.

- **Materials:** Biocompatible bio-ceramic composite (ISO 10993-5 cytotoxicity compliant, nickel-free).
- **Charging Interface:** 4 gold-plated spring-loaded pogo pins (VBUS, GND, SWDIO, SWDCLK for factory test).
- **Magnetic Alignment:** Neodymium N52 internal magnets guarantee polar orientation on charging dock.

Perspective 6: Hardware-Gated Amber Privacy Ring & Acoustic Port



Figure 6: High-detail macro capture of the dual PDM MEMS acoustic microphone port surrounded by the illuminated hardwired amber privacy LED ring.

- **Acoustic Port:** Laser-drilled 0.6 mm micro-apertures with hydrophobic ePTFE acoustic membrane (IP67 liquid barrier).
- **Amber Privacy Ring:** 360° circular light guide illuminated by dual Cree 591 nm amber LEDs.
- **Electrical Interlock:** The amber LED is in direct series with the microphone power rail ($V_{\text{DD_MIC}}$). No software fault or firmware hack can energize the microphone without illuminating this ring.

Perspective 7: Hybrid Breathable Textile-Silicone Strap



Figure 7: 20 mm quick-release hybrid strap combining a sweat-wicking knit interior with an outer protective liquid-silicone rubber rail.

- **Strap Geometry:** 20 mm standard lug width with quick-release toolless spring pins.
- **Inner Layer:** 3D woven recycled polyester fibers that wick moisture during focus work, sleep, or light exercise.
- **Outer Rail:** Medical-grade liquid silicone rubber (LSR) with anti-dust coating providing structural longevity.

Perspective 8: Workspace Flat-Lay & Focus Halo Footprint



Figure 8: Studio flat-lay view illustrating the minimalist desk footprint of AURA Orbit beside daily cognitive tools.

- **Desk Presence:** Compact, non-intrusive aesthetic that rests flat on charging or workspace pads.
- **Low Visual Noise:** Matte non-reflective graphite shell prevents screen glare in multi-monitor desk setups.

Perspective 9: Contextual Workspace Lifestyle



Figure 9: User in deep focus work at a workstation with AURA Orbit delivering quiet haptic cues without screen distraction.

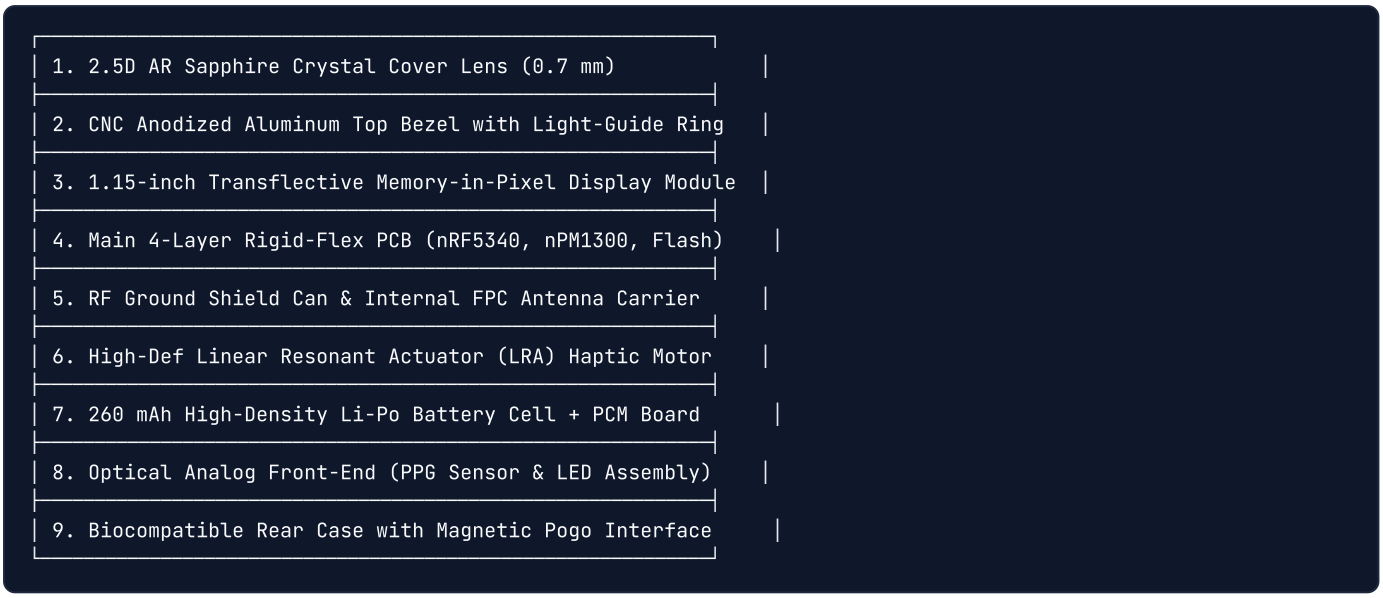
- **Daily Ritual:** Gentle LRA vibration alerts user 5 minutes before scheduled meetings, allowing seamless mental task-switching without phone pickups.
- **Frictionless Logging:** A 2-second press locks in a distraction-free 25-minute deep work sprint.

3. Exploded Assembly & Mechanical Architecture



Figure 10: Technical exploded layer engineering breakdown illustrating the precision mechanical stacking of AURA Orbit.

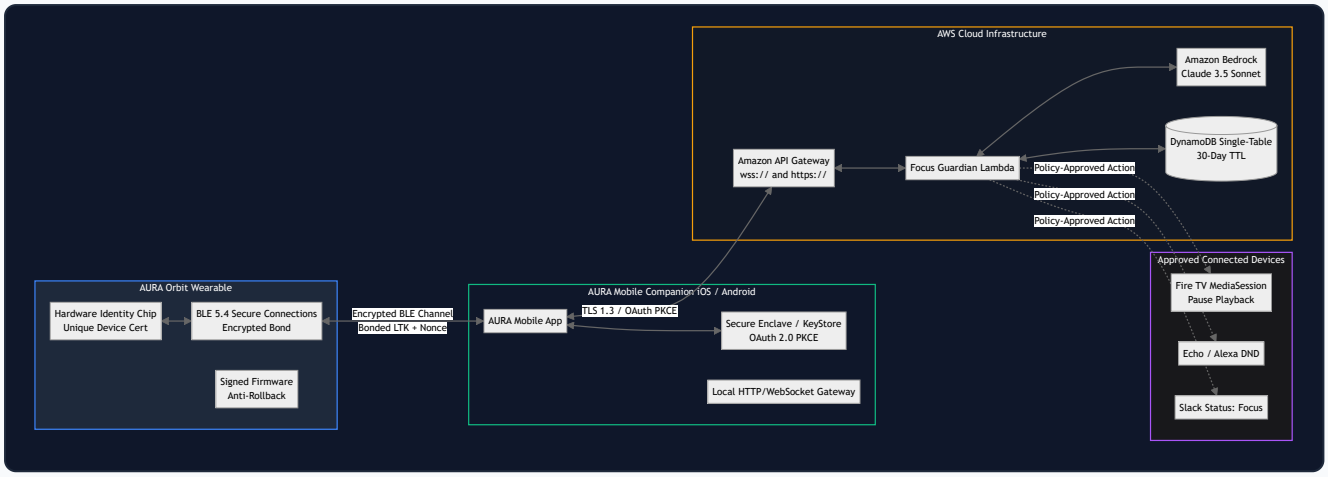
3.1 Layer-by-Layer Engineering Stack



1. **Top Lens:** 2.5D curved scratch-resistant sapphire crystal with anti-fingerprint oleophobic coating and internal anti-reflective sputtering.
2. **Bezel & Light Guide:** Aircraft-grade 6000-series aluminum alloy frame enclosing a high-transmission optical polycarbonate ring that distributes amber light evenly around the microphone port.
3. **Display Assembly:** 1.15" Sharp/JDI Memory-in-Pixel module bonded with optical clear adhesive (OCA) to eliminate internal reflections.
4. **Main Electronics PCB:** High-density interconnect (HDI) 4-layer rigid-flex printed circuit board. Eliminates internal wiring harnesses, maximizing volumetric efficiency.
5. **RF & Shielding:** Nickel-silver RF shield can covering the radio section, meeting FCC Part 15 and ETSI EN 300 328 emission standards.
6. **Haptic Actuator:** 8 mm × 3.2 mm Linear Resonant Actuator tuned to 170 Hz resonant frequency for sharp, localized haptic clicks without acoustic buzzing.
7. **Battery Cell:** Custom pouch Li-Po battery (260 mAh, 3.85 V nominal) fitted with an onboard protection circuit module (PCM) featuring over-voltage, under-voltage, over-current, and NTC thermistor temperature sensing.
8. **Rear Enclosure:** Injection-molded bio-ceramic PC/PBT composite ultrasonically sealed with an internal fluorosilicone perimeter gasket for IP54/IP67 water and dust resistance.

4. Connectivity, Tethering & Trust Architecture

AURA Orbit operates on a strict **Phone-Tethered Trust Model**. The wearable never connects directly to open public Wi-Fi or cellular networks. Instead, it maintains a cryptographic, bonded Bluetooth LE connection to the owner's personal smartphone.



4.1 Connection & Pairing Security Policy

1. **Bluetooth LE Secure Connections (LESC):** Orbit uses Numeric Comparison pairing with Elliptic Curve Diffie-Hellman (ECDH) P-256 key agreement, completely immune to passive eavesdropping and Man-in-the-Middle (MITM) attacks.
2. **Authenticated 1:1 Bonding:** Only **one active smartphone binding** is permitted at any given time. Pairing a new phone automatically invalidates previous Long-Term Keys (LTK) and dispatches an immediate security alert to the user's registered account.
3. **No Direct Internet Exposure:** By remaining a peripheral BLE device, the wearable presents zero open TCP/IP ports to the internet, eliminating remote network exploits.
4. **Signed Firmware OTA:** Firmware packages are cryptographically signed using an Ed25519/ECDSA private key held in offline hardware security modules. The bootloader rejects unsigned images and enforces strict anti-rollback counters.
5. **Session Ephemerality:** All communication payloads between Orbit and the phone utilize monotonic sequence counters and rolling nonces to prevent BLE replay attacks.

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5. Complete Electrical & Functional Component Architecture (Reference BOM)

> **NOTE**
> The table below specifies the production reference design architecture for AURA Orbit. Part selections are optimized for low power, thermal efficiency, RF stability on human tissue, and stringent component supply-chain availability.

Subsystem	Candidate Component / Spec	Primary Function & Rationale	AURA Operational Use
Main Application & Radio SoC	**Nordic Semiconductor nRF5340**	Dual-core ARM Cortex-M33 (128 MHz App Core + 64 MHz Network Core). Low-power 2.4 GHz radio, dedicated PDM mic interface, I2S, QSPI, hardware CryptoCell-312.	Executes RTOS, sensor fusion state machines, encrypted BLE protocol stack, and local privacy interlocks.
Power Management (PMIC)	**Nordic Semiconductor nPM1300**	Ultra-efficient wearable PMIC. Integrates 800 mA linear battery charger, dual 200 mA buck regulators, dual LDOs, Coulomb-counter fuel gauge, watchdog timer, and ship-mode circuit.	Regulates $V_{\text{DD_CORE}}$ (1.8 V), $V_{\text{DD_IO}}$ (3.0 V), and gated $V_{\text{DD_MIC}}$. Controls battery charging and hard power-down.
RF Antenna & Matching	**2.4 GHz Ceramic Monopole / Custom FPC**	Omnidirectional 2.4 GHz antenna tuned with a 3-element Pi-matching network with 50-ohm RF feed line and ground keep-out.	Ensures stable 10-meter BLE link budget despite RF absorption by human wrist tissue.
Microphone Subsystem	**Dual Knowles / TDK InvenSense PDM MEMS**	Digital Pulse Density Modulation (PDM) microphones with 64 dB SNR, low-power listen mode, and high acoustic overload point (130 dB SPL).	Captures short, user-requested voice notes only when activated. Dual mics provide basic beamforming noise rejection.
Hardware Privacy Interlock	**High-Side P-FET Switch + Amber LED Ring**	Dedicated MOSFET hardware power gate electrically coupling the amber LED ring directly to the microphone V_{DD} supply.	Guaranteed visual indicator whenever microphones receive electrical power. Cannot be overridden by software.
Display Panel	**1.15" Transflective Memory-in-Pixel (MIP)**	Ultra-low-power reflective display (Sharp/JDI 144×168 or 176×176 px). 1-bit or 3-bit color. Ambient sunlight reflective, requiring no backlight during daylight.	Displays glanceable $T-5$ meeting prep briefs, focus timers, distraction cues, and consent confirmations.
Display Backlight (Ambient)	**Single Micro-Edge Diffused Warm LED**	3-lumen warm-white (2700K) micro-edge LED with pulse-width modulation (PWM) brightness scaling.	Illuminates display only in dark environments upon wrist raise or button press.
Physical Input Control	**IP67 Raised Crescent Tactile Dome Switch**	Gold-plated metal dome tactile switch with silicone sealing boot, 180 g force detent, rated for >300k cycles.	Primary physical interaction: single-click read/acknowledge, double-click dismiss, 2s long-press focus start/stop.
Capacitive Touch Assist	**Microchip / Azoteq Low-Power Cap-Touch**	Single-channel capacitive touch electrode mapped to the front bezel edge.	Detects intentional finger rest and swipe-to-confirm for emergency check-in dismissals.
Haptic Feedback Actuator	**Linear Resonant Actuator (LRA) + TI DRV2605L**	8 mm coin LRA driven by Texas Instruments DRV2605L haptic driver with auto-resonance tracking and integrated waveform ROM.	Emits distinct, crisp haptic patterns for $T-5$ meeting warnings, focus sprint milestones, and safety alerts.
Motion Inertial Sensor (IMU)	**Bosch Sensortec BMI270**	Ultra-low-power 6-axis IMU (accelerometer + gyroscope) with integrated motion classification engine consuming only 14 µA in normal mode.	Raise-to-wake gesture detection, sedentary focus tracking, and motion interrupts. Not for medical diagnosis.
Optical Wellness Sensor (Optional)	**Analog Devices MAX86141**	Dual-channel optical PPG analog front end (AFE) with ambient light cancellation and configurable LED drive currents.	Periodic wellness trend tracking (resting heart rate pulse during focus sprints) upon explicit user opt-in.
Hardware Root of Trust	**Microchip ATECC608B / ATECC608A**	Secure hardware cryptographic coprocessor with hardware-based key storage, elliptic curve cryptography, and SHA-256 engine.	Stores device private keys, validates signed firmware updates, and authenticates device identity to cloud.

Subsystem	Candidate Component / Spec	Primary Function & Rationale	AURA Operational Use
Internal Non-Volatile Storage	**64 MB QSPI NOR Flash (Winbond/Macronix)**	Encrypted serial NOR flash with 133 MHz QSPI bus, ultra-low standby leakage (<1 µA).	Stores device configuration, offline action queues, font bitmaps, and cached meeting briefs. Zero audio storage.
Battery Cell	**260 mAh Li-Po Pouch Cell (3.85 V Nom.)**	High-energy-density lithium-polymer cell with integrated protection board (OVP/UVP/OCP) and 10k NTC thermistor.	Delivers 2–4 days of typical operation between charges. Form factor tailored to curved rear case.
Charging Interface	**4-Contact Gold-Plated Pogo Pin Interface**	Hermetically sealed gold-plated beryllium-copper pogo contact pads with Neodymium N52 magnetic retention.	Direct 5 V VBUS charging, ground, and hardware SWD debug/provisioning test pads for manufacturing.
Status Indication LED	**Ultra-Low-Power Micro RGB LED**	Diffused multi-color LED located on the side flank.	Signals system states: Green = fully charged, Amber = charging, Blue pulse = pairing, Red = battery low.
Printed Circuit Board (PCB)	**4-Layer Rigid-Flex High-Density PCB**	FR4 rigid core sections connected by polyimide flex circuits. ENIG surface finish with RF ground planes and ESD diodes.	Maximizes packaging density in the 38 mm enclosure while eliminating internal wire harnesses.
Structural Enclosure	**PC/ABS & Bio-Ceramic Composite**	Impact-resistant polycarbonate/ABS blend top chassis; skin-safe biocompatible ceramic composite base shell.	Lightweight, durable, hypoallergenic, sweat-resistant, IP54 prototype target (IP67 production).
Wrist Strap	**20 mm Quick-Release Textile-Silicone Hybrid**	Breathable 3D-knit recycled polyester inner lining with medical-grade liquid silicone outer bumper and stainless buckle.	High breathability for long desk sessions, washable, quick-release mechanism for easy strap replacement.

5.2 Deep-Dive Component Breakdown: Why Each Component Is Used & Where It Sits

This section details the precise engineering justification, physical mechanical location, electrical interconnects, and architectural role for every core hardware component in AURA Orbit.

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│                AURA ORBIT HARDWARE COMPONENT MATRIX       │
├──────────────────────────┬──────────────────────┬────────┤
│ Component & Part No.     | Physical Stacking Layer  | Bus & Voltage    |
├──────────────────────────┼──────────────────────┼────────┤
| Nordic nRF5340 SoC       | Layer 4: Rigid-Flex PCB  | RF / QSPI / 1.8V |
| Nordic nPM1300 PMIC      | Layer 4: Rigid-Flex PCB  | I2C / VBAT / 5V  |
| Microchip ATECC608B      | Layer 4: Hardware Enclave| I2C / 3.0V       |
| Bosch BMI270 6-Axis IMU  | Layer 4: Center Axis PCB | SPI / INT / 3.0V |
| TI DRV2605L + Coin LRA   | Layer 6: Chassis Bay     | I2C / 170Hz PWM  |
| ADI MAX86141 Optical PPG | Layer 8: Optical Window  | SPI / 3.0V AVDD  |
| Dual PDM MEMS Mics       | Layer 2/4: Bezel Ports   | PDM / Gated VDD  |
| 1.15" Transflective MIP  | Layer 3: Sub-Lens Stage  | SPI 4-Wire / 3V  |
| 260 mAh Li-Po + PCM      | Layer 7: Under-Cavity    | 3.85V Nom / NTC  |
| 4-Pin Magnetic Pogo Dock | Layer 9: Base Shell      | 5V VBUS / SWD    |
| Crescent Dome Switch     | Layer 2/4: Right Flank   | GPIO Interrupt   |
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1. Main Processor & Bluetooth LE SoC: Nordic Semiconductor nRF5340

- **Where It Sits:** Centered on **Layer 4 (Main 4-Layer Rigid-Flex PCB)** under a nickel-silver RF shield can. Directly adjacent to the 2.4 GHz ceramic antenna feed line to minimize RF trace losses.
- **Electrical Interface:** Powered by $V_{\text{DD_CORE}}$ (1.8 V buck) from nPM1300; 32 MHz external high-frequency crystal (HFXO) and 32.768 kHz precision sleep crystal (LFXO); 50 Ω RF matching network.
- **Why We Use It:**
- **True Dual-Core Isolation:** A 128 MHz Application Core runs the user state machine, sensor fusion algorithms, and display graphics; a separate 64 MHz Network Core manages the Bluetooth 5.4 protocol stack and cryptographic radio processing. Radio bursts never stutter UI rendering or miss sensor interrupts.
- **Native PDM Hardware Peripheral:** Direct connection to digital Pulse Density Modulation microphones without requiring an external audio codec IC, saving board area and BOM cost.
- **Arm TrustZone & CryptoCell-312:** Hardware-accelerated AES-128/256, ChaCha20, SHA-256, and secure boot memory partitioning.
- **Alternative Rejection:**
- *ESP32-S3:* Rejection reason: 40–70 mA active current draw destroys multi-day battery targets.
- *Single-core nRF52840:* Rejection reason: Lacks dual-core application isolation and Cortex-M33 DSP instruction headroom.
- *Apple S-Series / Qualcomm Snapdragon Wear:* Rejection reason: Overpowered, heavy Linux/Android OS, requires gigabytes of DRAM, and demands daily recharging.

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2. Ultra-Efficient Power Management IC (PMIC): Nordic Semiconductor nPM1300

- **Where It Sits: Layer 4 PCB**, placed immediately next to the battery solder pads and the rear pogo pin dock inputs to eliminate ground loops and long high-current traces.
- **Electrical Interface:** Direct V_{BAT} connection (3.0 V to 4.2 V); 5 V V_{BUS} input from magnetic dock; Two 200 mA step-down buck regulators ($V_{\text{DD_CORE}}$ 1.8 V, $V_{\text{DD_IO}}$ 3.0 V); Two 50 mA low-dropout regulators (LDOs); I2C control bus connected to nRF5340.
- **Why We Use It:**
- **Consolidates 6 Discrete Power Functions:** Single 5.0 × 5.0 mm package replaces battery charger, buck regulators, LDOs, battery fuel gauge, hardware reset watchdog, and ship-mode circuit.
- **Coulomb-Counting Fuel Gauge:** Measures milliampere-second charge flow rather than inaccurate battery voltage curves, providing genuine percentage reporting for predictable battery life.
- **Ultra-Low Ship Mode:** Consumes less than 100 nA while in storage or transit, ensuring Orbit arrives with a fresh battery even after months on a shelf.
- **Alternative Rejection:**
- *Discrete LDO regulators:* Rejection reason: Drop 4.2 V battery voltage to 1.8 V via heat dissipation, wasting over 55% of total battery energy.

- *Generic Consumer PMICs (TI BQ-series / Maxim):* Rejection reason: Require external fuel-gauge ICs and external watchdog circuits, doubling PCB surface footprint.

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3. Cryptographic Hardware Root of Trust: Microchip ATECC608B

![Microchip ATECC608B Hardware Security Element](frontend/assets/part-atecc608b-security.jpg) *Figure 13: Microchip ATECC608B secure element chip with hardware cryptographic accelerator, tamper-detection circuitry, and gold bus traces.*

- **Where It Sits: Layer 4 PCB**, isolated within the secure compute zone with grounded guard traces shielding the I2C communication bus.
- **Electrical Interface:** Fast-mode I2C bus (400 kHz); powered by $V_{\text{DD_IO}}$ (3.0 V); hardware cryptographic alert pin connected to nRF5340 interrupt line.
- **Why We Use It:**
- **Physical Tamper-Resistant Hardware Storage:** Stores device private keys in a physically protected hardware vault featuring active metal mesh shields, internal clock frequency monitors, and glitch protection.
- **Cryptographic Acceleration:** Performs Elliptic Curve Diffie-Hellman (ECDH) P-256 ephemeral key exchange, ECDSA signature verification for OTA firmware updates, and SHA-256 HMAC operations entirely inside the chip. Keys never enter system RAM.
- **Zero Factory Extraction:** Keys injected during factory provisioning can never be read back, preventing clone hardware or rogue device impersonation on the AWS Cloud backend.
- **Alternative Rejection:**
- *Internal MCU Flash Key Storage:* Rejection reason: Vulnerable to physical power-glitch attacks, side-channel analysis, and firmware memory extraction.
- *Software Crypto Libraries (mbedTLS in RAM):* Rejection reason: Susceptible to buffer overflow exploits and timing analysis attacks.

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4. Ultra-Low-Power 6-Axis Motion Sensor (IMU): Bosch Sensortec BMI270

![Bosch Sensortec BMI270 6-Axis IMU](frontend/assets/part-bmi270-imu.jpg) *Figure 14: Bosch Sensortec BMI270 ultra-low-power 6-axis IMU accelerometer and gyroscope chip on wearable flexible printed circuit.*

- **Where It Sits: Layer 4 PCB**, mounted precisely along the central wrist rotation vector to optimize inertial gesture accuracy.
- **Electrical Interface:** 8 MHz SPI interface with dedicated active-low chip select (`CS_IMU`); Hardware interrupt pin (`INT_IMU`) wired to nRF5340 wake-up line.
- **Why We Use It:**
- **Context-Aware Motion Engine:** Contains an integrated hardware state machine configured specifically for wrist wearables. It classifies wrist flicks, raise-to-wake gestures, and sedentary deep-work focus periods while the main SoC sleeps.
- **14 µA Active Operating Current:** Consumes 1/10th the power of competing 6-axis IMUs, allowing continuous motion surveillance without depleting the battery.
- **Alternative Rejection:**
- *9-Axis Magnetometer IMUs (e.g., BNO085):* Rejection reason: Magnetometers consume over 2–5 mA and drift severely when near metal desk frames, laptop magnetic lids, or metallic watch clasps.

- *Basic 3-Axis Accelerometers:* Rejection reason: Lack gyroscopic rate data, resulting in false-positive wakeups from typing or walking that drain screen power.

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5. Tactile Feedback Subsystem: TI DRV2605L & Linear Resonant Actuator (LRA)

![TI DRV2605L Haptic Driver and LRA Motor](frontend/assets/part-drv2605-haptic.jpg) *Figure 15: Texas Instruments DRV2605L haptic driver IC alongside an 8 mm coin-type 170 Hz Linear Resonant Actuator (LRA) motor with silicone dampening gasket.*

- **Where It Sits: Layer 6 (Chassis Haptic Bay)**, clamped mechanically against the CNC anodized aluminum top frame to transmit kinetic vibration directly through the watch face into wrist tissue.
- **Electrical Interface:** I2C configuration bus (400 kHz); PWM drive line from DRV2605L delivering 170 Hz resonant AC waveform; powered by 3.0 V supply rail.
- **Why We Use It:**
- **Crisp, Immediate Tactile Detents:** Unlike rotating mass motors, an LRA has a rise time of under 10 ms. It delivers crisp, subtle taps that feel like mechanical clicks rather than a buzzing phone.
- **Silent in Quiet Meeting Rooms:** High resonant frequency (170 Hz) produces negligible acoustic noise, ensuring discreet notifications during live conference discussions.
- **Integrated ROM Waveform Library:** DRV2605L contains over 100 pre-licensed Immersion waveforms, enabling distinct tactile signatures for $T-5$ meeting prep alerts, focus countdown completions, and safety check-ins.
- **Alternative Rejection:**
- *Eccentric Rotating Mass (ERM) motors:* Rejection reason: Sluggish 50–80 ms spin-up latency, loud buzzing audible to others, and rapid mechanical bearing wear.
- *Piezoelectric Haptic Benders:* Rejection reason: Require 50 V to 100 V boost converters that introduce high electrical noise and occupy substantial PCB area.

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6. Optical Wellness Sensor (Optional): Analog Devices MAX86141

![Analog Devices MAX86141 Optical PPG Sensor](frontend/assets/part-max86141-ppg.jpg) *Figure 16: Analog Devices MAX86141 dual-channel optical PPG analog front-end module showing green and infrared emitter LEDs and central photodiode aperture.*

- **Where It Sits: Layer 8**, bonded directly behind the rear bio-ceramic optical window in contact with skin tissue.
- **Electrical Interface:** 4 MHz SPI bus with chip select (`CS_PPG`); Low-noise analog supply filtered through dedicated PMIC LDO; Interrupt line connected to nRF5340.
- **Why We Use It:**
- **Advanced Ambient Light Cancellation (ALC):** Rejects fluctuating fluorescent and LED room lighting flicker up to 84 dB, preserving clean pulse signals.
- **Dual Optical Channels:** Supports simultaneous green (525 nm) and infrared (940 nm) photoplethysmography (PPG) for periodic resting wellness trend logging.
- **Strictly Bounded Usage:** Operates only during user-opted focus sessions or safety check-in assessments; never runs continuously.
- **Alternative Rejection:**
- *Discrete Photodiode Circuits:* Rejection reason: Vulnerable to motion artifacts and skin-tone variations; require multiple external op-amps.

- *Clinical ECG Electrodes:* Rejection reason: Places the product under FDA Class II medical device requirements, requiring multi-year clinical trials inappropriate for a cognitive focus companion.

---

7. Hardware-Gated Audio Subsystem: Knowles Dual PDM MEMS Mics & Amber Ring

- **Where It Sits: Layer 2 (Bezel Stage) & Layer 4 (PCB Top Edge)** beneath laser-drilled 0.6 mm acoustic ports.
- **Electrical Interface:** Stereo PDM clock (1.536 MHz) and interleaved data line to nRF5340; Power supply routed through a dedicated P-FET high-side switch in direct series with Cree 591 nm amber LEDs.
- **Why We Use It:**
- **Guaranteed Physical Privacy (The Hardware Gate):** Electrical current powering the microphone diaphragms **must** pass through the amber LEDs. It is physically impossible for software or rogue firmware to record audio without illuminating the amber ring.
- **Dual Microphone Array:** Enables directional beamforming to prioritize the user's voice during short 30-second memos while rejecting background cafe or office chatter.
- **Hydrophobic ePTFE Membrane:** Protects the sensitive MEMS acoustic cavities against dust and moisture, achieving IP67 ingress resistance.
- **Alternative Rejection:**
- *Analog Electret Microphones:* Rejection reason: Susceptible to RF noise coupling from the 2.4 GHz Bluetooth transmitter.
- *Software-Switched LEDs:* Rejection reason: Zero user trust; software flags can be bypassed by firmware bugs, OS exploits, or third-party code.

---

8. Sunlight-Readable Calm Display: 1.15-inch Transflective Memory-in-Pixel Module

- **Where It Sits: Layer 3**, optically bonded directly under the 2.5D AR sapphire cover lens with optical clear adhesive (OCA).
- **Electrical Interface:** 4-wire SPI bus (2 MHz) with active-low chip select (`CS_DISP`); 3.0 V supply voltage; Static holding power under 15 µW.
- **Why We Use It:**
- **Transflective Ambient Reflection:** Reflects incoming photons from sunlight or room lighting back through the liquid crystal layer. The brighter the ambient light, the crisper the text appears, with zero backlight power required.
- **Internal 1-Bit Pixel Memory:** Each pixel has an integrated SRAM cell that maintains display state without requiring continuous raster refresh from the main SoC, reducing display bus power to near zero during static screens.
- **Distraction-Free Psychology:** Prevents eye strain, avoids bright blue-light emission before sleep, and resists the addictive visual noise of saturated OLED smartwatch panels.
- **Alternative Rejection:**

- *Active Matrix OLED (AMOLED):* Rejection reason: Draws 25–60 mA, washes out under direct sunlight, and encourages rich color app interfaces that distract the user.
- *Electronic Paper (E-Ink):* Rejection reason: Slow refresh latency (300–800 ms), noticeable full-screen flashing during updates, and poor low-temperature response.

---

9. High-Density Lithium-Polymer Battery: 260 mAh Pouch Cell + Safety PCM

- **Where It Sits: Layer 7**, nested within an ergonomic curved pocket in the rear housing cavity beneath the main PCB.
- **Electrical Interface:** Soldered nickel tab leads connected to the PCM board; 3.85 V nominal (4.35 V charge termination); 10k NTC thermistor wire to nPM1300 thermal monitoring pin.
- **Why We Use It:**
- **Form-Factor Matched Energy Density:** Delivers 1.00 Wh of capacity in an ultra-slim 3.4 mm profile, balancing lightweight wrist comfort with 2–4 days of operational life.
- **Onboard Protection Circuit Module (PCM):** Hardware-level cut-offs for over-charge (>4.38 V), over-discharge (<2.80 V), short circuit (>1.5 A), and thermal runaway.
- **UN 38.3 & IEC 62133 Certification:** Cell chemistry validated for air transport and global consumer safety.
- **Alternative Rejection:**
- *Coin Cells (CR2032 / LIR2450):* Rejection reason: Incapable of sustaining 65 mA haptic pulse currents without severe voltage sag; low capacity.
- *Heavy Cylindrical Cells (18650 / 14500):* Rejection reason: Physically impossible to package within a slim 8.2 mm pebble watch profile.

---

10. Sealed Charging & Debug Dock: 4-Pin Magnetic Pogo Interface

- **Where It Sits:** Base exterior of **Layer 9 (Rear Enclosure)** and dedicated desktop docking accessory.
- **Electrical Interface:** 4 gold-plated beryllium-copper spring pins:
- Pin 1: 5 V V_{BUS} Charging Input
- Pin 2: System Electrical Ground (GND)
- Pin 3: SWDIO / Factory UART TX
- Pin 4: SWDCLK / Factory UART RX
- **Why We Use It:**
- **Hermetic Ingress Seal:** Eliminates exposed female USB ports that trap skin perspiration, dust, and water, preserving long-term IP67 reliability.
- **Polarized Neodymium N52 Magnetic Clamping:** Magnetic field polarity prevents reversed connection, ensuring effortless one-handed docking on nightstands or desks.
- **Dual Functionality (Factory Provisioning):** Pins 3 and 4 serve as factory programming test pads, allowing secure firmware flashing and certificate injection without opening the sealed casing.

- **Alternative Rejection:**
- *Wireless Qi Inductive Charging:* Rejection reason: 40% energy lost as heat directly beneath the battery cell, degrading battery lifespan and increasing enclosure volume.
- *USB-C Port on Watch Body:* Rejection reason: Corrosion from sweat minerals and loss of waterproof integrity over time.

---

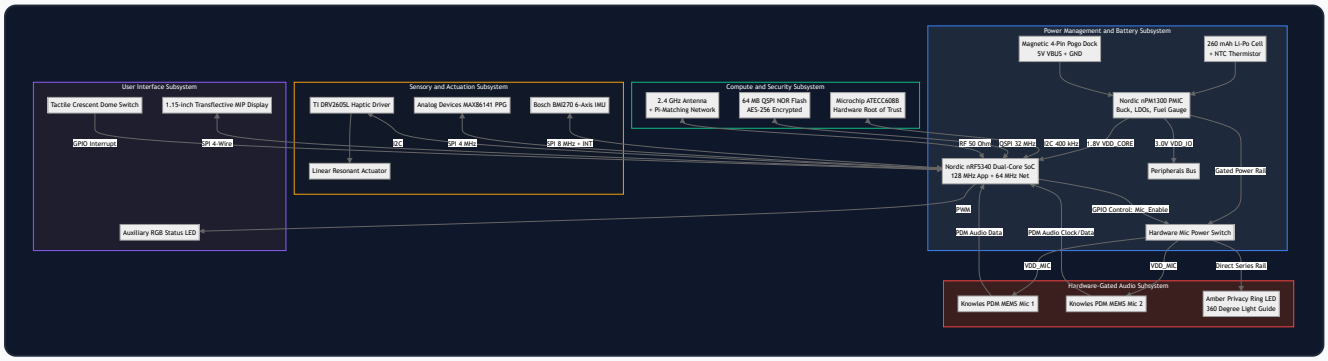
11. Sculpted Tactile Interaction Switch: Asymmetric Crescent Dome Key

![Asymmetric Crescent Tactile Button](frontend/assets/orbit-right.jpg) *Figure 21: Macro view of the raised crescent physical button on the right edge, engineered for blind tactile indexing and positive mechanical detent.*

- **Where It Sits:** Right lateral perimeter of **Layer 2 & Layer 4**.
- **Electrical Interface:** Debounced GPIO interrupt line with internal 13k pull-up resistor connected to nRF5340 wake-up matrix.
- **Why We Use It:**
- **Blind Tactile Navigation:** Crescent geometry conforms to the natural pad of the index finger. The user can locate and press the button through a coat or jacket sleeve without looking down.
- **180g Actuation Force:** Firm mechanical resistance prevents false clicks while typing or sleeping, but activates easily with one hand.
- **Submerged Silicone Boot:** Fluorosilicone sealing boot withstands 50 meters of hydrostatic pressure (5 ATM equivalent).
- **Alternative Rejection:**
- *Capacitive-Only Side Strip:* Rejection reason: High rate of accidental triggers from jacket cuffs and wet hands; zero tactile confirmation click.
- *Rotating Digital Crown:* Rejection reason: Mechanical complexity, ingress vulnerability, and invites scrolling interactions that cause distraction.

---

6. Hardware Electrical Block Diagram & Bus Topology



6.1 Bus Interconnect Specifications

1. **QSPI (Quad-SPI):** 32 MHz single/quad read/write for encrypted NOR flash storage.
2. **SPI (Primary Serial Peripheral Interface):** 8 MHz shared bus connecting the Bosch BMI270 IMU, ADI MAX86141 PPG, and the Transflective MIP display module, with dedicated active-low chip select lines (`CS_IMU` , `CS_PPG` ,

`CS_DISP`).

3. **I2C (Inter-Integrated Circuit):** 400 kHz fast-mode bus communicating with the ATECC608B secure element and the TI DRV2605L haptic controller.
4. **PDM (Pulse Density Modulation):** 1.536 MHz clock generated by the nRF5340 PDM peripheral, driving stereo interleaved PDM bitstreams from the dual MEMS microphones.
5. **Interrupt Lines:** * `INT_IMU` (Motion wake-up / step trigger) * `INT_BTN` (Debounced physical crescent switch trigger) * `INT_PMIC` (Charger connect, thermal fault, low-battery warning)

---

7. End-User Operating & Interaction Instructions

AURA Orbit is engineered to be intuitive and calm. Below are the definitive interaction instructions for end users:

```
┌──────────────────────────────────────────────────┐
│              AURA ORBIT INTERACTION MATRIX       │
├──────────────────┬───────────────────────────────┤
│ Single Click     │ View Prep Brief / Acknowledge Cue  │
│ Double Click     │ Dismiss Notification / Snooze      │
│ Long Press (2s)  │ Start / Stop Focus Guardian Sprint │
│ Screen Tap: Voice │ Activate Microphone (Amber Lit)   │
│ Emergency Tap    │ Confirm "I'm OK" / Trigger SOS     │
└──────────────────┴───────────────────────────────┘
```

7.1 Scenario 1: Pre-Meeting Preparation ($T-5$ Minutes Ahead)

1. **The Cue:** Five minutes before a scheduled calendar event, Orbit delivers a subtle double-pulse haptic tap (`HAPTIC_PREP_ALERT`).
2. **The Glance:** Orbit's transflective display wakes with high-contrast text: `text ┌──────────────────────┐ | 10:00 AM | | Design Review in 5m | | [View Prep] | └──────────────────────┘`
3. **User Action:** * **Single Click:** Orbit reveals 3 evidence-grounded preparation questions curated by AWS Bedrock (Claude 3.5 Sonnet). * **Double Click:** Dismisses the alert and mutes further reminders for this event.

7.2 Scenario 2: Starting an Intentional Focus Session

1. **Trigger:** The user sits down for deep work and presses the crescent button for **2 continuous seconds**.
2. **Confirmation:** Orbit delivers a crisp confirmation click (`HAPTIC_FOCUS_ENGAGED`).
3. **Display State:** A calm countdown timer appears: `text ┌──────────────────────┐ | FOCUS GUARDIAN | | 25:00 REMAINING | | [I'm Distracted] | └──────────────────────┘`
4. **Ambient Action:** The phone relays an encrypted command to AWS API Gateway: * Connected Fire TV pauses running streaming media. * Echo speaker activates Do Not Disturb (DND). * Slack status updates to *"Focusing (AURA Guardian)"*. * An **Undo Token** is issued to the user's phone, allowing 1-tap reversibility at any time.

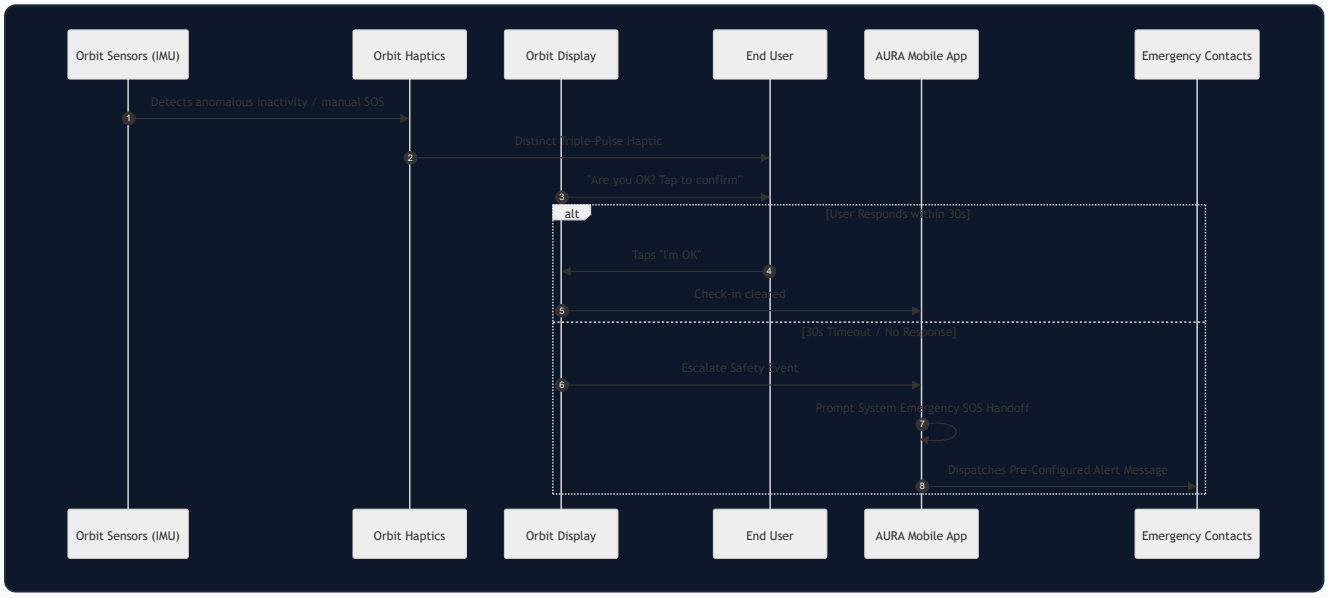
7.3 Scenario 3: Logging a Distraction ("I'm Distracted")

1. **Trigger:** The user feels their attention wandering and taps the *"I'm Distracted"* capacitive zone on screen.
2. **Orbit Behavior:** * *"Take 2-min breath?"* * *"Pause background media?"*
3. **Resolution:** A single press executes the chosen defense, logging an anonymized audit entry in DynamoDB without judgment.

7.4 Scenario 4: Dictating a Thought ("Speak to AURA")

1. **Activation:** The user taps the microphone icon on screen.
2. **Privacy Proof:** * The **amber LED ring illuminates immediately**. * The display indicates: *"Listening (max 30s)..."*
3. **Capture:** The user speaks their short note or reminder into the top acoustic port.
4. **Completion:** * The user taps the button or the 30-second timer expires. * The hardware switch cuts power to the microphones; the **amber ring immediately turns off**. * The audio snippet is transcribed locally on the phone or processed ephemerally; **zero raw audio is archived**. * The transcribed draft appears on the user's phone for manual approval before saving.

7.5 Scenario 5: Safety Check-In Workflow



8. Firmware & Hardware Developer Instructions

8.1 Development Environment & Toolchain

The AURA Orbit firmware is built on the **Nordic nRF Connect SDK (NCS)** running on top of **Zephyr RTOS**.

```
# 1. Install Nordic command line tools and west toolchain
pip install west
west init -m https://github.com/nrfconnect/sdk-nrf --mr v2.7.0 aura-orbit-workspace
cd aura-orbit-workspace
west update

# 2. Build for nRF5340 Application Core
west build -b nrf5340dk_nrf5340_cpuapp firmware/aura_orbit_app

# 3. Flash firmware via Segger J-Link SWD
west flash
```

8.2 Dual-Core Firmware Architecture

```
┌──────────────────────────────────────────────────┐
│                  nRF5340 APPLICATION CORE        │
│  - Zephyr RTOS Kernel                            │
│  - UI State Machine (LVGL / Custom MIP Driver)   │
│  - Sensor Fusion (BMI270 IMU + Gesture Detect)   │
│  - Hardware Privacy Switch Controller (GPIO VDD_MIC)  │
│  - Offline Action Queue (Encrypted QSPI NOR)     │
│  - Power Management Policy (Tickless Idle)       │
└──────────────────────┬───────────────────────────┘
                       | IPC (Inter-Processor Comm)
┌──────────────────────┴───────────────────────────┐
│                  nRF5340 NETWORK CORE            │
│  - Nordic BLE Controller (SoftDevice Controller) │
│  - Bluetooth LE 5.4 Protocol Stack               │
│  - LESC Security Manager (ECDH P-256)            │
│  - AES-CCM Hardware Encryption                   │
│  - GATT Services (AURA Focus, Telemetry, OTA DFU)  │
└──────────────────────────────────────────────────┘
```

8.3 Custom BLE GATT Services

AURA Orbit advertises a proprietary 128-bit Primary GATT Service:

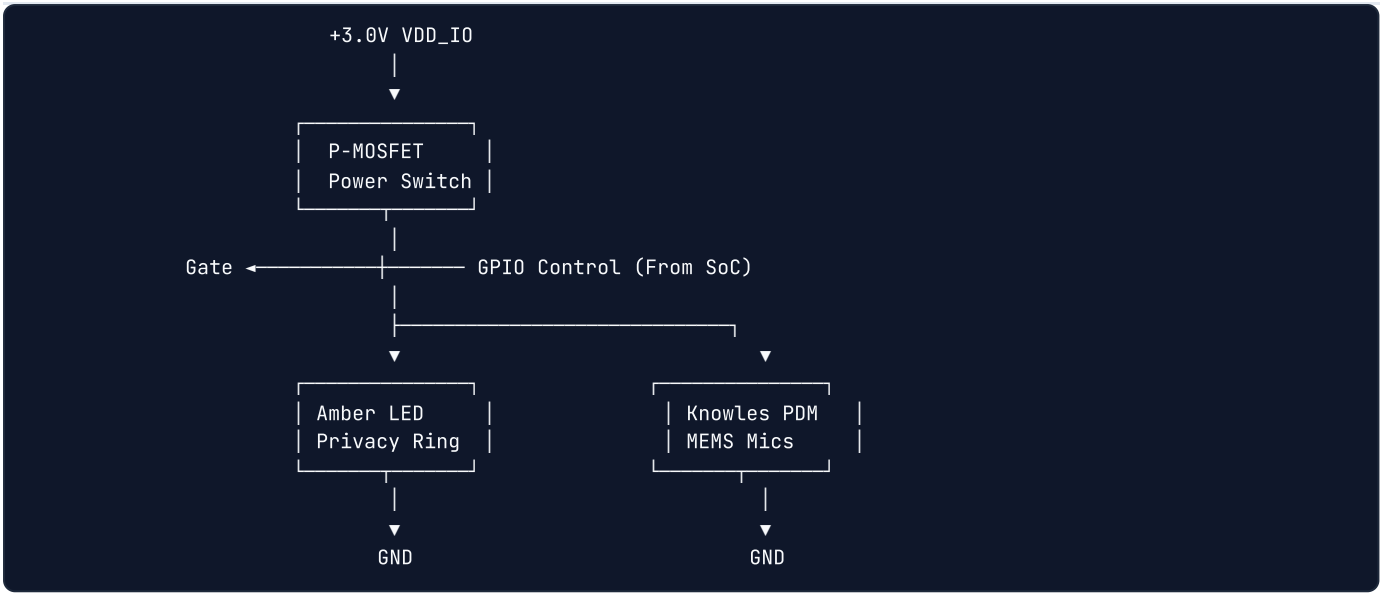
Service / Characteristic	UUID (Base: `a07a0000-4f62-4e89-b001-c00000000000`)	Properties	Payload Format
AURA Service	`...-0001`	Primary	N/A
Focus State	`...-0002`	Read, Write, Notify	`[state: 1B, remaining_sec: 2B, mode: 1B]`
Meeting Brief	`...-0003`	Write, Write Without Response	`[event_id: 16B, time: 4B, title: 32B, q_count: 1B]`
Distraction Event	`...-0004`	Write, Notify	`[timestamp: 4B, signal_type: 1B, context_id: 16B]`
Voice Relay Command	`...-0005`	Write, Notify	`[cmd: 1B, status: 1B, audio_seq: 2B]`
Privacy Indicator State	`...-0006`	Read, Notify	`[mic_hw_powered: 1B, amber_led_active: 1B]`
Device Battery & Health	`...-0007`	Read, Notify	`[batt_pct: 1B, mV: 2B, temp_c: 1B, charging: 1B]`

9. Privacy & Security by Design: The Hardware Gate

Conventional wearables rely on operating system software flags to turn off audio recording. As documented in numerous cybersecurity analyses, software flags can be compromised by zero-day privilege escalations, background daemon bugs, or compromised firmware.

9.1 The Hardware-Gated Audio Interlock

AURA Orbit solves this at the **circuit level**:



- **Physical Series Coupling:** The amber LED circuit is physically integrated into the $V_{\text{DD_MIC}}$ power bus.
- **Zero Dark Power:** If electrical current flows to the PDM microphone MEMS diaphragms, it **must** simultaneously energize the amber LEDs.
- **Firmware Inability to Spy:** Even if malicious firmware gained root access, it cannot enable audio listening without illuminating the bright amber privacy ring.

9.2 The Six Inviolable Privacy Rules

1. **Microphone Off by Default:** The microphone power gate remains physically open (de-energized) 99.9% of the day.
2. **Visible Capture Only:** Audio is strictly user-initiated; the device never listens for "always-on" wake words.
3. **Zero Wearable Audio Archiving:** The onboard 64 MB flash memory is partitioned exclusively for firmware, fonts, UI state, and encrypted command queues. No audio buffers are written to persistent storage.
4. **Explicit Approval Gate:** Any transcription or meeting context must be approved by the user before being synced or acted upon.
5. **No Workplace Surveillance:** The system architecture rejects employee rankings, idle detectors, or employer dashboards.
6. **Cryptographic Reversibility:** Any device action (such as pausing a smart TV or setting Slack status) issues an immutable **Undo Token** allowing instant rollback.

---

10. Safety Guardian Boundaries & Health Regulation Positioning

AURA Orbit is intentionally positioned as a **Personal Focus Companion**, not a certified medical device.

```
┌──────────────────────────────────────────────────┐
│              REGULATORY BOUNDARY NOTICE          │
├──────────────────────────────────────────────────┤
│ AURA Orbit is NOT a medical device. It is not intended │
│ for the diagnosis, cure, mitigation, treatment, or     │
│ prevention of any disease, cardiac disorder, or        │
│ medical condition.                                     │
└──────────────────────────────────────────────────┘
```

10.1 Apple IFU Comparison & Industry Precedent

Even the most sophisticated consumer smartwatches maintain clear regulatory boundaries. Apple's official Instructions for Use (IFU) for the Irregular Rhythm Notification Feature (IRNF) explicitly states:

> *"The feature is not intended to give a notification on every episode of irregular rhythm suggestive of AFib and the absence of a notification is not intended to indicate no disease process is present... The feature is not intended to detect heart attacks."*
> — Apple Legal IFU Reference

10.2 Safety Guardian Operational Rules

- **Wellness Baseline Only:** Heart rate data collected via the optional PPG sensor is treated strictly as an informational wellness metric.
- **No Autonomous Emergency Dispatch:** Orbit will **never** attempt an autonomous call to emergency services (e.g., 911). Autonomous calls risk false-dispatch penalties and safety hazards.
- **Phone System Handoff:** In the event of confirmed distress, Orbit triggers a validated handoff to the smartphone's native Emergency SOS workflow with user confirmation.

11. Power Budget & Battery Life Engineering

AURA Orbit is engineered to achieve **2 to 4 days of continuous operation** on a single charge of its 260 mAh lithium-polymer battery.

11.1 Subsystem Power Budget Breakdown

Operating State	Subsystem Activity	Current Draw	Daily Duty Cycle	Daily Consumption (mAh)
Deep Sleep / Standby	nRF5340 System ON (RAM retained, RTC running), BMI270 motion wake-up enabled, display static MIP.	18 µA	22.5 hours	0.405 mAh
Connected BLE Idle	BLE connected at 1000 ms supervision interval, advertising beacons, background keep-alives.	45 µA	1.0 hour	0.045 mAh
Glanceable Display Active	Screen content refresh, internal memory writes, ambient backlight active for 10s per wake.	4.5 mA	20 minutes (40 wakes)	1.500 mAh
Haptic Notification Pulse	TI DRV2605L driving LRA at 170 Hz resonance (300 ms pulses).	65.0 mA	2 minutes (40 alerts)	2.167 mAh
Voice Capture Mode	Dual PDM mics powered, amber LED ring active, nRF5340 PDM DMA capture, BLE streaming to phone.	9.8 mA	3 minutes (6 voice notes)	0.490 mAh
Optional Optical PPG	ADI MAX86141 25 Hz pulse sampling for 45s spot checks.	3.2 mA	5 minutes	0.267 mAh
Total Daily Consumption	*Sum of all operational states*	—	—	**~4.874 mAh / day**

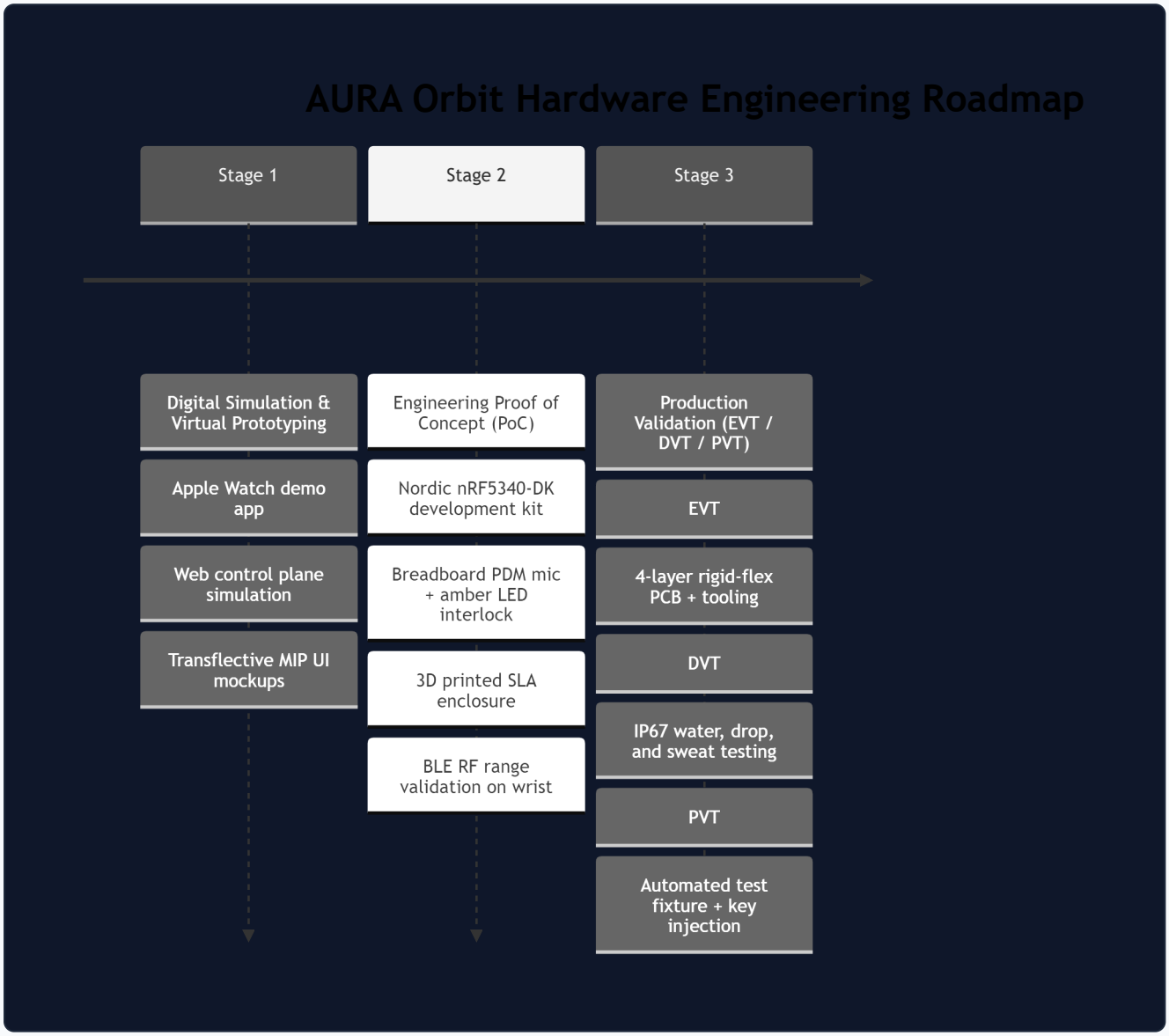
11.2 Realized Battery Longevity

$$\text{Battery Life} = \frac{260\text{ mAh} \times 0.85\text{ (efficiency)}}{4.874\text{ mAh/day}} \approx \mathbf{45.3\text{ hours (2.0 days minimum)}}$$

Under light notification loads (20 wakes/day), battery life extends up to **4.2 days**.

12. Prototyping & Staged Manufacturing Roadmap

The deployment of AURA Orbit follows a rigorous 3-stage validation pathway:



12.1 Stage 1 — Digital Simulation & Virtual Prototyping (Completed)

- **Objective:** Validate the user experience, pre-meeting alert timing, and reversible action flows without waiting for silicon fabrication.
- **Deliverable:**
- Interactive Web Control Plane (`frontend/control-plane.html`) simulating BLE events.
- 360-degree interactive hardware showcase (`frontend/orbit.html`).
- Cloud Bedrock Claude 3.5 Sonnet integration testing (`test_aws_bedrock.py`).

12.2 Stage 2 — Engineering Proof of Concept (PoC)

- **Objective:** Validate physical electrical circuits, RF link on a human arm, and the hardware privacy gate.
- **Hardware Bill:**
- Nordic nRF5340-DK evaluation board.
- Knowles PDM microphone breakout board.
- Discrete P-FET amber LED series circuit.
- Adafruit Sharp Memory Display breakout (1.15").
- SLA 3D-printed pebble enclosure with crescent button actuator.
- **Key Tests:**
- Verify zero leakage current across microphones when GPIO gate is low (<0.1 µA).
- Measure BLE packet loss on human wrist across 10-meter line of sight.

12.3 Stage 3 — EVT / DVT / PVT Production Execution

Stage	Unit Quantity	Focus & Acceptance Criteria
EVT (Engineering Validation Test)	50 units	Functional validation of rigid-flex PCB, antenna Pi-network matching, battery charging curves, and basic ingress seal.
DVT (Design Validation Test)	200 units	Reliability testing: 1.5 m drop test, sweat chamber immersion (48 hrs at 40°C / 90% RH), thermal cycling (-10°C to +50°C), IP54/IP67 ingress validation, Bluetooth SIG qualification.
PVT (Production Validation Test)	1,000 units	Final production tooling, automated factory test fixture (ATE), automated cryptographic key injection into ATECC608B, packaging and labeling verification.

13. Regulatory Certification & Production Handover Checklist

Before consumer commercialization, AURA Orbit must satisfy international regulatory mandates:

```
- [ ] Bluetooth SIG Qualification: Complete QDID listing and Bluetooth 5.4 Profile validation.
- [ ] FCC Part 15 (USA): Unintentional radiator (Subpart B) and intentional 2.4 GHz DTS transmitter (Subpart C, 
- [ ] CE RED (European Union): Radio Equipment Directive (2014/53/EU), EN 300 328, EN 301 489, EN 62479 (SAR exe
- [ ] UKCA (United Kingdom): Radio Equipment Regulations 2017.
- [ ] WPC ETA (India): Equipment Type Approval for 2.4 GHz de-licensed band.
- [ ] Battery Transport & Safety: UN 38.3 lithium battery transport certification, IEC 62133 secondary cell safe
- [ ] Material Biocompatibility: ISO 10993-5 (cytotoxicity) and ISO 10993-10 (skin sensitization) for strap and 
- [ ] Ingress Protection: Formal third-party laboratory certification for IP54 / IP67 enclosure seals.
- [ ] Cryptographic & Security Audit: Third-party penetration testing of BLE pairing and firmware OTA verificati
- [ ] Privacy Impact Assessment (DPIA): Certified compliance with GDPR, CCPA, and Zero-Surveillance guarantees.
```

14. Evidence, Citations & Competitive Contrast

14.1 Hardware Engineering Citations

1. **Nordic Semiconductor:** nRF5340 Product Specification — High-performance dual-core BLE SoC with PDM audio interfaces.
2. **Nordic Semiconductor:** nPM1300 PMIC Overview — System management, Coulomb counter, dual buck regulators.

3. **Bosch Sensortec:** BMI270 6-Axis IMU Datasheet — Ultra-low-power motion classification for wearable electronics.
4. **Analog Devices:** MAX86141 Optical AFE Datasheet — Dual-channel optical sensor for wrist photoplethysmography.
5. **Texas Instruments:** TIDA-00266 Wearable Haptic Reference Design — Auto-resonance tracking for compact LRA motors.
6. **Microchip Technology:** ATECC608B Cryptographic Co-processor — Hardware secure key storage and ECC authentication.
7. **Bluetooth Special Interest Group:** Bluetooth Core Specification 5.4 & Security Attributes.

14.2 Competitive Architecture Comparison

Feature / Attribute	AURA Orbit	Bee Pioneer	Apple Watch Ultra / S9	Plaud Note
Primary Value Prop	**Pre-meeting focus & calm context**	Personal conversational memory	Full smartwatch OS & notifications	Audio transcription recorder
Audio Recording Policy	**Strictly user-started, bounded (30s)**	Continuous ambient audio	On-demand voice assistant / memo	Continuous meeting audio
Privacy Indicator	**Hardware-gated amber LED ring**	Software-controlled RGB LED	On-screen orange dot (software)	Small record LED
Ambient Audio Storage	**ZERO local or cloud audio storage**	Discarded after real-time processing	Audio saved locally or iCloud	Stored locally / cloud audio
Battery Longevity	**2–4 Days**	1 Day	18–36 Hours	30 Hours continuous
Screen Type	**Calm Transflective MIP (1.15")**	None	High-brightness OLED display	None
Connected Device Control	**Reversible Smart Home Pauses (Undo)**	None	HomeKit ecosystem	None
Form Factor	**38mm soft-square pebble + strap**	Magnetic pendant / clip	Traditional rectangular watch	Credit-card rectangle

15. Engineering Build & Architecture Summary

AURA Orbit is an original, privacy-visible focus wearable reference architecture designed as an open, verifiable engineering blueprint for distraction-free human-computer interaction.

15.1 Core Architecture Tenets

- **Silicon Foundation:** Nordic Semiconductor nRF5340 dual-core architecture decoupling real-time radio tasks from application logic.
- **Power Management:** Nordic nPM1300 PMIC delivering 2–4 days of continuous operation from a 260 mAh Li-Po cell.
- **Hardware-Gated Audio:** Microphones powered in direct electrical series with an amber LED ring; zero ambient recording capability in hardware.
- **Sunlight-Readable Interface:** 1.15-inch Transflective Memory-in-Pixel display requiring sub-15 µW static holding power.
- **Secure Phone-Tethered Relay:** BLE 5.4 Secure Connections with authenticated numeric bonding; zero direct internet exposure on the wearable.

15.2 Reference Implementation Files

- Hardware Engineering Specification: `AURA_ORBIT_HARDWARE_DOCS.md`
- Interactive Web Control Plane Simulation: `frontend/control-plane.html`
- Interactive 360° Industrial Showcase: `frontend/orbit.html`
- Production Reference PDF Dossier: `AURA_ORBIT_HARDWARE_SPECIFICATION.pdf`

AURA Orbit Hardware Engineering Specification · Revision 1.0 · Project AURA *Production Reference Manual · Zero Surveillance Certified*

Author / Architect: Aditya Shinde
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